Supporting Material: Scale-Bridging Techno-Economic, Microclimatic, and Open-Quantum Assessment of Semi-Transparent Organic Agrivoltaics for Sub-Saharan Perennial Crops

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Abstract

This Supporting Material documents the complete optical transfer-matrix formulation, microscopic 14-site LHCII exciton–vibrational Hamiltonians and MesoHOPS 1.8.0 convergence checks, hourly ERA5 meteorological dataset extraction, tri-crop Farquhar–von Caemmerer–Berry (FvCB)–Medlyn–FAO-56 governing equations, the Penja white pepper (*poivre blanc de Penja*, IGP) versus generic black pepper parameterization, cooperative battery dispatch and life-cycle accounting rules, and supplementary scenario tables.

S1. Optical transfer-matrix formalism and spectral calibration

Multilayer optical propagation across the semi-transparent organic photovoltaic (ST-OPV) stack—glass substrate (1.1 mm, incoherent) / ITO anode (100 nm) / ZnO ETL (35 nm) / PM6:Y6 (1:1.2) photoactive layer ($d_{\mathrm{act}} \in \{50, 60, 75, 90, 105, 120\}$ nm) / MoO_3 HTL (10 nm) / ultrathin Ag electrode ($d_{\mathrm{Ag}} \in \{8, 10, 12, 14, 16\}$ nm) / MoO_3 dielectric capping layer ($d_{\mathrm{cap}} \in \{0, 20, 35, 50\}$ nm)—is evaluated using the 2×2 Abeles characteristic transfer-matrix method at 1 nm spectral resolution ($\lambda \in [300, 1100]$ nm). Unpolarized irradiance responses average transverse-electric (TE) and transverse-magnetic (TM) modes across incident angles $\theta \in [0°, 80°]$; the integration stops at 80° because the Abeles stack becomes ill-conditioned at grazing incidence, and the neglected solid-angle weight above 80° is $\cos^2 80° \approx 0.03$ of the normal-incidence weight:

$$T_{\mathrm{OPV}}^{\mathrm{beam}}(\lambda, \theta) = \frac{1}{2}\left[T_{\mathrm{TE}}(\lambda, \theta) + T_{\mathrm{TM}}(\lambda, \theta)\right], \qquad T_{\mathrm{OPV}}^{\mathrm{diff}}(\lambda) = 2\int_0^{\pi/2} T_{\mathrm{OPV}}^{\mathrm{beam}}(\lambda, \theta) \sin\theta \cos\theta \, \mathrm{d}\theta. \quad \text{(S1)}$$

Complex refractive indices $n(\lambda) + i\kappa(\lambda)$ use published variable-angle spectroscopic ellipsometry datasets for PM6:Y6, D18:Y6, and ITO alongside dispersive Johnson–Christy Ag optical constants, avoiding constant-index artifacts. Across all 120 Pareto-swept stacks, local energy conservation satisfies $\max|R(\lambda) + T(\lambda) + A_{\mathrm{tot}}(\lambda) - 1| = 4.44 \times 10^{-16}$. An opaque reference cell ($d_{\mathrm{act}} = 75$ nm, $d_{\mathrm{Ag}} = 100$ nm) reproduces the experimental PM6:Y6 benchmark ($J_{\mathrm{sc}} = 25.701\,\mathrm{mA\,cm^{-2}}$, PCE = 15.501 % vs. $26.36\,\mathrm{mA\,cm^{-2}}$ and 16.27 % reported by Shen et al. [1]).

Photosynthetically active transmittance (APT, 400 to 700 nm), extended PAR (ePAR, 400 to 750 nm), light utilization efficiency (LUE = PCE × APT), and photomorphogenic red-to-far-red shift (Δ(R:FR) comparing $[655, 665]$ nm to $[725, 735]$ nm) follow AM1.5G photon-flux

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weighting (ASTM G-173-03). The 24-design Pareto frontier yields $\mathrm{LUE}_{\max} = 4.424\,\%$ (APT $\in [14.54\,\%, 52.06\,\%]$), with the focal stack APT44 ($d_{\mathrm{act}} = 60\,\mathrm{nm}, d_{\mathrm{Ag}} = 12\,\mathrm{nm}, d_{\mathrm{cap}} = 35\,\mathrm{nm}$) delivering APT = 44.12 % and PCE = 9.93 %.

S2. Microscopic 14-site LHCII exciton–vibrational model and CW NESS bounds

S2.1. Three independent 14-site Hamiltonians and dipoles

To eliminate single-Hamiltonian bias, three monomeric LHCII 14-pigment Frenkel-exciton Hamiltonians (sites Chl b_{601}–a_{614}) are constructed directly from primary literature: H1 takes the Redfield–Förster pigment-site energies and couplings of Novoderezhkin et al. [2] paired with point-dipole transition geometry from Liu et al. (PDB 1RWT); H2 adopts the electrostatic charge-density coupling site energies and Poisson–TrEsp couplings of Müh et al. [3] (PDB 2BHW); and H3 implements the MMSIC-MD parameter set of Zhu, Higashi, and Saito [4] and Kim et al. [5], with a symmetric Hermitian coupling matrix ($\max|H_{ij} - H_{ji}| = 0.0\,\mathrm{cm}^{-1}$) led by Chl a_{611}–a_{612} at $+129.0\,\mathrm{cm}^{-1}$ and Chl a_{604}–b_{606} at $+114.0\,\mathrm{cm}^{-1}$. Transition dipole strengths ($\mu^2_{\mathrm{Chl}\,a} = 21.0\,\mathrm{D}^2$, $\mu^2_{\mathrm{Chl}\,b} = 14.7\,\mathrm{D}^2$) preserve exact non-orthogonality along the crystallographic $\mathrm{N_B}$–$\mathrm{N_D}$ molecular axes (dimer 610–611 in PDB 1RWT: $\cos\theta = +0.932$, $\kappa = -1.812$), enabling partial-secular Fano interference testing. Each chlorophyll couples to a Drude–Lorentz overdamped protein bath ($\lambda_D = 50.87\,\mathrm{cm}^{-1}$ matching the Renger spectral density integral [5], bounded over $[35.0, 140.0]\,\mathrm{cm}^{-1}$, $\gamma_D = 50.0\,\mathrm{cm}^{-1}$) plus high-frequency intramolecular modes (340 and $740\,\mathrm{cm}^{-1}$) at $T = 295\,\mathrm{K}$ across $N = 200$ static Gaussian disorder realizations (FWHM $= 110\,\mathrm{cm}^{-1}$).

S2.2. Continuous-wave non-equilibrium steady state vs. pulsed MesoHOPS

Under continuous-wave (CW) solar illumination filtered by $T_{\mathrm{OPV}}(\lambda)$, broadband photon-pumping operators act without inter-mode temporal phase correlation [6, 7]. Solving the Liouville steady state $\mathcal{L}\rho_{\mathrm{ss}} = 0$ compares the full Bloch–Redfield partial-secular tensor (η_{PS}) against its diagonal secular Pauli counterpart (η_{Pauli}) via

$$\Delta\eta_{\mathrm{NESS}} = \eta_{\mathrm{PS}} - \eta_{\mathrm{Pauli}}, \qquad \eta = \frac{k_{\mathrm{RC}}\,\mathrm{Tr}(P_{\mathrm{term}}\rho_{\mathrm{ss}})}{k_{\mathrm{RC}}\,\mathrm{Tr}(P_{\mathrm{term}}\rho_{\mathrm{ss}}) + k_{\mathrm{NR}}\,\mathrm{Tr}(\rho_{\mathrm{ss}}) + k_{\mathrm{NPQ}}\,\mathrm{Tr}(P_{\mathrm{qE}}\rho_{\mathrm{ss}})}, \tag{S2}$$

with open-RC trapping $k_{\mathrm{RC}} = 0.005\,\mathrm{ps}^{-1}$ (swept over $[0.0033, 0.5]\,\mathrm{ps}^{-1}$), nonradiative decay $k_{\mathrm{NR}} = 0.25\,\mathrm{ns}^{-1}$, and q_E photoprotective quenching $k_{\mathrm{NPQ}} \in \{0, 1.5\}\,\mathrm{ns}^{-1}$. The bound is reported as a magnitude, but the crop model receives a signed perturbation sampled over the stated interval (baseline $[-0.02, +0.10]\,\%$, exact-HEOM envelope $[-0.21, +0.21]\,\%$), so no sign of the coherence shift is assumed. A 4-site exact QuTiP-BoFiN HEOM cross-check [8] verifies $|\eta_{\mathrm{HEOM}} - \eta_{\mathrm{Pauli}}| = 0.002\,10$ (H2) and $0.000\,106$ (H3).

Separately, full non-Markovian 14-site MesoHOPS 1.8.0 trajectories ($N = 48$ disorder seeds, 2000 fs propagation, max trace-norm deviation $\leq 1.11 \times 10^{-15}$) confirm that coherent transient wavepacket dynamics under pulsed initial conditions deviate markedly from secular kinetics (terminal Chl a_{610}–a_{612} MAE = 0.0198 on H2–DL, 0.2017 on H3–DL, and 0.2200 on H3–DL+Vib_{340}, where $340\,\mathrm{cm}^{-1}$ vibronic dynamics reach $p_{\mathrm{term}} = 0.0835$ vs. 0.4035 Pauli), yet vanish under CW incoherent sunlight ($|\Delta\eta_{\mathrm{NESS}}| \leq 3.370 \times 10^{-5}$).

S3. ERA5 microclimate, tri-crop ecophysiology, and Penja white pepper parameterization

Hourly meteorological drivers (8760 h, 2023 UTC) are extracted from Open-Meteo ERA5 reanalysis [9] for Foumbot (1044 m, Arabica coffee), Njombe-Penja (273 m, Penja white pepper), and Djuttitsa (1400 m, highland tea). Diffuse fractions are obtained from ERA5 direct/diffuse decomposition (annual averages 0.365, 0.461, and 0.412; wet season 0.416 to 0.502;

dry season 0.260 to 0.360). Sub-canopy radiation solves 2D row-geometry shadow projection with diffuse sky view factor F_{sky} (generating low-PPFD shading fractions $F_{\mathrm{lp}} \in [0.277, 0.544]$ under ST-OPV depending on GCR). Coupled leaf and module temperatures resolve hourly sensible, radiative, and latent energy balances (residual $\leq 1.569 \times 10^{-5}\,\mathrm{W\,m^{-2}}$). Crop assimilation couples the Farquhar–von Caemmerer–Berry (FvCB) model [10] with Medlyn optimal stomatal conductance [11] and daily FAO-56 two-bucket water stress (AWC = 160 mm) [12], plus calibrated irradiance photoinhibition thresholds ($I_{\mathrm{PI}} = 900\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ for *Piper nigrum*, $1200\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ for *Camellia sinensis*). Exact Shapley causal decomposition across all $2^4 = 16$ coalition subsets guarantees machine-precision additivity (residual $\leq 7.105 \times 10^{-15}\,\%$).

S3.1. Penja white pepper (poivre blanc de Penja) vs. black pepper parameterization

Table S1 documents why Penja white pepper parameters avoid conflation with whole-dried black pepper.

Table S1: Agronomic, post-harvest processing, and market parameterization separating Protected Geographical Indication (PGI) Penja white pepper from generic black pepper.

Parameter	**Model Value (White)**	**Black Pepper Contrast**	**Agronomic / Market Basis**
Clean dry yield	$800\,\mathrm{kg\,ha^{-1}\,yr^{-1}}$	$1100\,\mathrm{kg\,ha^{-1}\,yr^{-1}}$ equivalent	White kernel recovery is ∼24 % of red-ripe fresh berries vs. ∼34 % whole-dried black pepper.
Baseline farm-gate price	$5000\,\mathrm{XAF\,kg^{-1}}$	1800 to $2800\,\mathrm{XAF\,kg^{-1}}$	Cameroon Njombe-Penja peeled white kernel price (3500 to $6000\,\mathrm{XAF\,kg^{-1}}$ range); PGI export tier $= 14\,000\,\mathrm{XAF\,kg^{-1}}$.
Post-harvest electric load	$45.0\,\mathrm{kWh\,t^{-1}}$ (85 % day)	10 to $15\,\mathrm{kWh\,t^{-1}}$ (sun drying)	Sized for spring-water retting circulation pumps (7 to 10 d), mechanical decortication/washing, and fan-assisted solar drying.
Gas-exchange ($V_{c\max 25}, J_{\max 25}$)	44.6, $65.0\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$	Species-shared (*Piper nigrum* L.)	Calibrated in steady-state FvCB against Oliveira et al. [13] and Ferreira et al. [14].

S4. Extended scenario results (S_0–S_5)

S4.1. Sobol robustness checks across exact-HEOM quantum bounds and zero-carbon pricing

To test sensitivity under stricter counterfactual bounds, all $N = 10\,000$ Saltelli sequences (140 000 model evaluations per crop under an identical quasi-Monte Carlo sequence) are re-evaluated across four uncertainty-set variants (Table S3): (1) the baseline policy ensemble ($\Delta\eta_{\mathrm{NESS}} \in [-0.02, +0.10]\,\%$, carbon price $[0, 25]\,\mathrm{USD\,tCO_2e^{-1}}$); (2) the exact-HEOM quantum envelope ($\Delta\eta_{\mathrm{NESS}} \in [-0.21, +0.21]\,\%$, matching $|\eta_{\mathrm{HEOM}} - \eta_{\mathrm{Pauli}}| = 2.10 \times 10^{-3}$); (3) strict zero-carbon Monte Carlo ($0\,\mathrm{USD\,tCO_2e^{-1}}$); and (4) the combined exact-HEOM and zero-carbon configuration. Even under the widest combined envelope, comparative quantum total-order Sobol sensitivity never exceeds 2.59×10^{-7} on ΔNPV or 9.01×10^{-8} on standalone $\mathrm{NPV_{OPV}}$, while exceedance odds remain bounded within 11.0 to 24.7 %. In contrast, classical macro-financial parameters govern standalone $\mathrm{NPV_{OPV}}$ (grid export tariff $S_{Ti} = 0.5176$ in coffee and

Table S2: Summary of focal agronomic, PV generation, and techno-economic outputs under grid-connected operation at the baseline 0 USD tCO_2e^{-1} carbon price and 0 % CAPEX subsidy. NPV and LCA use a 2 h LFP buffer for coffee/pepper and overnight withering BESS for tea in S_3(APT44, GCR65), S_5(APT44, GCR65, irrig.), and S_2(c-Si, GCR40), whereas S_3(APT56, GCR50) and S_3(APT52, GCR65) reflect daytime-only dispatch without storage.

Crop	**Scenario**	**Shade** (%)	ΔY_{ann} (%)	ΔY_{dry} (%)	ΔY_{wet} (%)	**PV Yield** (MWh ha^{-1} yr^{-1})	**LER** (–)	**NPV** (kUSD ha^{-1})	**LCA** (gCO_2e kWh^{-1})
Coffee	S3 (APT56, GCR50)	25.7	−1.21	+9.02	−6.96	717.0	1.265	213.0	33.99
Coffee	S3 (APT44, GCR65)	40.6	−7.08	+8.82	−15.88	1223.3	1.401	403.5	26.16
Coffee	S5 (APT44, GCR65, irrig.)	40.6	+0.22	+38.23	−15.88	1223.3	1.474	414.2	26.18
Coffee	S2 (c-Si, GCR40)	40.0	−6.38	+8.73	−14.92	1479.0	1.507	779.3	18.92
Penja Pepper	S3 (APT56, GCR50)	25.9	+7.27	+19.57	+1.53	611.7	1.350	205.7	39.84
Penja Pepper	S3 (APT52, GCR65)	36.0	+8.16	+27.56	+0.04	870.2	1.477	253.3	36.85
Penja Pepper	S3 (APT44, GCR65)	40.8	+7.43	+30.24	−1.36	1043.7	1.548	368.4	30.73
Penja Pepper	S5 (APT44, GCR65, irrig.)	40.8	+12.25	+49.68	−1.36	1043.7	1.596	376.5	30.74
Penja Pepper	S2 (c-Si, GCR40)	40.0	+6.95	+27.89	−0.78	1257.2	1.640	709.1	22.31
Highland Tea	S3 (APT56, GCR50)	25.8	+0.01	+11.26	−3.47	652.5	1.271	556.3	37.35
Highland Tea	S3 (APT44, GCR65)	40.7	−3.35	+15.98	−8.84	1113.3	1.429	860.8	32.66
Highland Tea	S5 (APT44, GCR65, irrig.)	40.7	+2.47	+41.38	−8.84	1113.3	1.487	899.8	32.91
Highland Tea	S2 (c-Si, GCR40)	40.0	−2.30	+16.56	−7.65	1374.4	1.548	1272.6	23.46

0.4263 in pepper; WACC $S_{Ti} = 0.4503$ and diesel price $S_{Ti} = 0.2817$ in highland tea), whereas hardware parameters ($C_{\text{mod}}^{\text{OPV}}$, T_{80}, and $C_{\text{BOS}}^{\text{c-Si}}$) govern comparative parity ΔNPV as summarized in Main Text Table 5.

Table S3: Robustness of Saltelli–Sobol exceedance probabilities $P(\text{NPV}_{\text{OPV}} > \text{NPV}_{\text{c-Si}})$ and quantum total-order sensitivity $S_{Ti}(\Delta\eta_{\text{NESS}})$ evaluated on ΔNPV across $N = 10\,000$ base samples.

Variant	**Crop**	$P(>$ c-Si) (%)	ΔNPV_{p50} (kUSD ha^{-1})	$S_{Ti}^{\text{Q}}(\Delta\text{NPV})$ (–)	$S_{Ti}^{\text{Q}}(\text{NPV}_{\text{OPV}})$ (–)	$S_{Ti}(C_{\text{mod}}^{\text{OPV}})$ (–)
Baseline	Coffee	16.75	−139.3	3.83×10^{-9}	1.16×10^{-9}	0.343
Baseline	Penja Pepper	22.99	−105.3	2.10×10^{-8}	7.15×10^{-9}	0.366
Baseline	Highland Tea	11.00	−176.0	9.85×10^{-9}	1.89×10^{-9}	0.323
HEOM (±0.21 %)	Coffee	16.74	−139.3	4.69×10^{-8}	1.42×10^{-8}	0.343
HEOM (±0.21 %)	Penja Pepper	22.97	−105.3	2.58×10^{-7}	8.76×10^{-8}	0.366
HEOM (±0.21 %)	Highland Tea	11.00	−176.0	1.21×10^{-7}	2.32×10^{-8}	0.323
Zero-Carbon	Coffee	18.51	−131.0	3.87×10^{-9}	1.20×10^{-9}	0.346
Zero-Carbon	Penja Pepper	24.71	−97.9	2.12×10^{-8}	7.35×10^{-9}	0.369
Zero-Carbon	Highland Tea	12.34	−166.5	9.95×10^{-9}	1.95×10^{-9}	0.327
HEOM + Zero-Carbon	Coffee	18.51	−131.0	4.74×10^{-8}	1.46×10^{-8}	0.346
HEOM + Zero-Carbon	Penja Pepper	24.72	−97.9	2.59×10^{-7}	9.01×10^{-8}	0.369
HEOM + Zero-Carbon	Highland Tea	12.33	−166.6	1.22×10^{-7}	2.39×10^{-8}	0.327

S4.2. Convergence and seed stability of the Saltelli–Sobol decomposition

The indices in Table 5 come from one scrambled sequence, so their numerical stability was tested rather than assumed. Repeating the identical decomposition on a resolution ladder ($N = 1250$, 2500, 5000, 10 000 base draws) and across three independent scrambles of the quasi-Monte Carlo sequence (`scripts/wp5c_sobol_convergence.py`) leaves the ranking of drivers unchanged in every case (Table S4). At the coarsest rung the three hardware total-order indices differ from their $N = 10\,000$ values by at most 0.007, and across scrambles at $N = 10\,000$ they span at most 0.005. The exceedance probability $P(\Delta\text{NPV} > 0)$ varies by at most 0.6 pp across scrambles, and the bounded quantum index moves only in its third significant digit. These ranges describe the sampling behaviour of the estimator: they are not confidence intervals for a population parameter, and they do not convert the deterministic reanalysis years of Section S6.1 into a probability sample. Parameters outside the twelve-parameter ensemble (soil water capacity, boundary-layer conductance, crop physiology) remain fixed at their literature values and were audited separately (Section S5).

Table S4: Seed and resolution stability of the comparative Sobol decomposition. Entries give the minimum–maximum range across three independent scrambles (seeds 42, 1234, 2024) at $N = 10\,000$ base draws; all three reported hardware drivers retain their rank order in every run. Ranges are sampling diagnostics, not confidence intervals.

Crop	$S_{Ti}(C_{\mathrm{mod}}^{\mathrm{OPV}})$ (–)	$S_{Ti}(T_{80})$ (–)	$S_{Ti}(C_{\mathrm{BOS}}^{\mathrm{c\text{-}Si}})$ (–)	$P(\Delta > 0)$ (%)	$S_{Ti}^{Q}(\Delta\mathrm{NPV})$ (–)
Arabica coffee	0.343–0.345	0.2466–0.2471	0.1593–0.1595	16.57–17.15	3.83×10^{-9}–3.84×10^{-9}
Penja white pepper	0.366–0.369	0.2595–0.2599	0.1701–0.1704	22.90–23.12	2.103×10^{-8}–2.107×10^{-8}
Highland tea	0.323–0.325	0.2671–0.2674	0.1500–0.1503	10.97–11.12	9.85×10^{-9}–9.87×10^{-9}

S5. Coupled structural ($AWC \times g_b$), meteorological stress, and seasonal monsoon-open louver audits

To quantify biophysical parameter uncertainty that enters upstream of the 12-parameter Sobol QMC ensemble (Section 5.3), `scripts/wp3b_structural_and_louver_audit.py` propagates the full 8760 h coupled FvCB–Medlyn–FAO-56–LWIR engine across two non-Sobol axes without altering baseline canonical datasets: (i) a coupled 3×3 factorial sweep of rainfed root-zone available water capacity AWC $\in \{120, 160, 220\}$ mm and leaf boundary-layer conductance $g_b \in \{0.6, 1.2, 2.5\}\,\mathrm{mol\,m^{-2}\,s^{-1}}$, and (ii) two physically consistent synthetic meteorological stress perturbations around the verified 2023 ERA5 forcing ("warm–dry": $\Delta T_{\mathrm{air}} = +1.5\,^{\circ}\mathrm{C}$, $\Delta\mathrm{RH} = -5\,\mathrm{pp}$, precipitation $\times 0.75$, GHI $\times 1.05$, $\Delta f_{\mathrm{diff}} = -0.05$; "cool–cloudy": $\Delta T_{\mathrm{air}} = -1.0\,^{\circ}\mathrm{C}$, $\Delta\mathrm{RH} = +5\,\mathrm{pp}$, precipitation $\times 1.25$, GHI $\times 0.90$, $\Delta f_{\mathrm{diff}} = +0.08$, recomputing e_a and ambient VPD via Tetens saturation). Each corner is referenced to an open-sun control (S_0) evaluated under identical (AWC, g_b) or climate forcing (Table S5). Across all three crops, wind-coupled boundary-layer conductance g_b dominates soil AWC by more than an order of magnitude: dense, calmer canopies ($g_b = 0.6\,\mathrm{mol\,m^{-2}\,s^{-1}}$) elevate open-sun leaf temperature and render agrivoltaic cooling strongly yield-positive (+5.20 to +29.17 %), whereas well-mixed windy canopies ($g_b = 2.5\,\mathrm{mol\,m^{-2}\,s^{-1}}$) keep open-sun leaves near ambient air temperature and shift shaded yields mildly light-limited (−13.40 to −1.88 %), while preserving narrow inter-technology parity (−1.14 to +2.41 pp).

We also quantify the counterfactual seasonal retraction design (S_6, Table S6) suggested qualitatively for cloudy wet months in Section 5.1, wherein panels on S_3 (PM6:Y6 ST-OPV, GCR = 0.65) and S_2 (c-Si, GCR = 0.40) retract edge-on to full open sky from June through October ($\mathrm{GCR}_h = 0$, $\tau_{\mathrm{PAR},h} = 1$) while remaining closed November–May. Re-evaluating the coupled 8760 h crop engine, factory/BESS dispatch, 20 yr DCF economics, and dual-FU ISO 14040 LCA shows that although June–October retraction rescues monsoon photosynthesis by +1.22 to +7.38 pp (S_3) and +0.98 to +6.98 pp (S_2), the associated 38.0 to 40.1 % loss in annual electricity revenue reduces 20 yr NPV by 349 596 to 784 209 USD ha^{-1} relative to fixed arrays (and by 357 696 to 826 855 USD ha^{-1} relative to S_5 dry-season drip irrigation), yielding zero (0.00 USD m^{-2}) capital headroom for motorized louver actuation wherever grid-connected power monetization anchors bankability.

S6. Weather-year and accepted-export sensitivity analyses

S6.1. ERA5 weather-year sensitivity, 2014–2023

The main analysis uses 2023 hourly ERA5 forcing. To assess interannual weather sensitivity with crop, optical, and economic parameters held fixed, we repeated the WP3 simulations for each year from 2014 to 2023 at the three modeled sites; leap years contribute 8784 hourly samples and the remaining years 8760; annual sums and low-PPFD fractions are normalised per year, so the 0.3 % longer leap-year basis does not propagate into the comparison. These deterministic reanalysis scenarios are neither independent random draws nor a probability sample,

Table S5: Coupled 3×3 (AWC × g_b) structural sensitivity and synthetic 2023 meteorological stress perturbations for PM6:Y6 ST-OPV (S_3, APT = 44 %, GCR = 0.65) versus opaque c-Si (S_2, GCR = 0.40). Each yield shift ΔY is referenced to open sun (S_0) under the same parameter or climate corner; $S_3 - S_2$ denotes percentage-point contrast (pp).

Condition / corner	AWC	g_b	Coffee (ΔY, %)			Pepper / Tea (ΔY, %)		
	(mm)	(mol m^{-2} s^{-1})	S_3	S_2	$S_3 - S_2$	S_3 (Pep / Tea)	S_2 (Pep / Tea)	$S_3 - S_2$
Low AWC, sheltered	120	0.6	+5.73	+5.97	−0.24	+25.86 / +5.20	+23.57 / +6.15	+2.29 / −0.95
Low AWC, baseline g_b	120	1.2	−7.25	−6.54	−0.71	+6.24 / −3.62	+5.78 / −2.56	+0.47 / −1.06
Low AWC, ventilated	120	2.5	−13.40	−12.44	−0.96	−4.42 / −8.21	−3.95 / −7.07	−0.47 / −1.14
Baseline AWC, sheltered	160	0.6	+5.97	+6.20	−0.23	+27.34 / +5.55	+25.00 / +6.50	+2.34 / −0.94
Canonical 2023 baseline	160	1.2	−7.08	−6.38	−0.70	+7.43 / −3.35	+6.95 / −2.30	+0.48 / −1.05
Baseline AWC, ventilated	160	2.5	−13.26	−12.30	−0.96	−3.46 / −7.98	−2.98 / −6.85	−0.47 / −1.13
High AWC, sheltered	220	0.6	+6.42	+6.63	−0.22	+29.17 / +6.13	+26.76 / +7.02	+2.41 / −0.89
High AWC, baseline g_b	220	1.2	−6.78	−6.09	−0.69	+8.87 / −2.90	+8.37 / −1.90	+0.50 / −1.00
High AWC, ventilated	220	2.5	−13.01	−12.06	−0.95	−2.34 / −7.61	−1.88 / −6.52	−0.47 / −1.09
Synthetic warm–dry stress	160	1.2	+1.28	+1.72	−0.44	+16.67 / +1.67	+15.29 / +2.65	+1.38 / −0.98
Synthetic cool–cloudy stress	160	1.2	−12.82	−11.91	−0.91	+0.74 / −7.77	+0.79 / −6.66	−0.05 / −1.10

Table S6: Quantified counterfactual performance of seasonal monsoon-open louvers (S_6, retracted June–October to full sky) relative to fixed rainfed arrays (S_3/S_2) and dry-season irrigated fixed arrays (S_5) under baseline 20 yr zero-subsidy economics.

Crop	Array	ΔY (%)		Rescue	PV (MWh ha^{-1} yr^{-1})		LCOE (USD kWh^{-1}) / CO_2e (gCO_2e kWh^{-1})		ΔNPV$_{S_6-\text{fix}}$
		Fixed	S_6	(pp)	Fixed	S_6	Fixed	S_6	(USD ha^{-1})
Coffee	ST-OPV (S_3)	−7.08	+0.30	+7.38	1223.3	738.8	0.0637 / 26.2	0.1056 / 43.3	−379 337
Coffee	c-Si (S_2)	−6.38	+0.60	+6.98	1479.0	890.8	0.0380 / 18.9	0.0631 / 31.4	−468 579
Pepper	ST-OPV (S_3)	+7.43	+8.65	+1.22	1043.7	627.3	0.0750 / 30.7	0.1248 / 51.1	−349 596
Pepper	c-Si (S_2)	+6.95	+7.93	+0.98	1257.2	752.8	0.0449 / 22.3	0.0750 / 37.3	−425 667
Tea	ST-OPV (S_3)	−3.35	+1.50	+4.85	1113.3	689.8	0.0886 / 32.7	0.1445 / 53.0	−678 246
Tea	c-Si (S_2)	−2.30	+1.97	+4.26	1374.4	849.2	0.0555 / 23.5	0.0909 / 38.2	−784 209

and they support neither confidence intervals nor field validation. The 2023 baseline outputs were kept separately.

Table S7: Focal ST-OPV modeled annual crop-yield change and shade-nearest OPV–c-Si yield contrast across 2014–2023 ERA5 forcing. Means and standard deviations are across ten deterministic weather years; positive-year counts are descriptive.

Crop	Mean ΔY_{OPV} (%)	SD (pp)	Range (%)	Positive years	Mean OPV–c-Si contrast (pp)	Positive contrast years
Arabica coffee	−8.22	1.11	−11.45 to −6.38	0/10	−0.74	0/10
Penja white pepper	+1.85	2.01	−1.43 to +7.60	7/10	+0.17	8/10
Highland tea	−3.54	0.97	−5.85 to −1.86	0/10	−1.11	0/10

For Penja white pepper the shade-nearest OPV–c-Si contrast spans −0.06 to +0.50 percentage points and therefore crosses zero. The positive-year counts describe this ten-year, fixed-parameter reanalysis set only and carry no probability of future outcomes. The exercise resolves weather-year variability alone; uncertainty in crop parameters, soil properties, future climate, and model structure remains outside it.

The focal design is approximately shade-matched to the c-Si reference and is not energy-matched. Across 2014–2023 the per-crop mean generation reached 81.2 to 82.8 % of the shade-nearest c-Si design's annual generation; the 2023 envelope across crops was 81.0 to 83.0 %. No tested equal-shade pair also matched annual electricity generation. Matching c-Si output would require lowering its ground-coverage ratio to 30 to 35 %, which alters the crop-light exposure the comparison is designed to hold. Crop light is therefore matched approximately, electrical service is not.

S6.2. Accepted-export accounting sensitivity

The full-export baseline accepts all otherwise-exportable electricity. In the sensitivity case the dispatch accepts 50 % of otherwise-exportable annual energy and curtails the remainder, with local consumption and each crop's selected battery configuration unchanged. The fraction is applied to annual energy and is neither an hourly interconnection limit nor a grid-availability model, and no local purchase agreement is assumed. NPV is reported per hectare of installed array under the modeled cooperative-load allocation.

Table S8: Modeled NPV for the focal ST-OPV scenario under full, 50 %-accepted, and zero grid export. Each crop uses its selected battery configuration. NPV is per installed-array hectare; export acceptance is a stylized annual energy constraint.

Crop	Selected battery	Full export	50 % accepted	No export
			NPV (USD ha^{-1})	
Arabica coffee	Small LFP, 2 h	403 543.53	11 895.55	−379 752.43
Penja white pepper	Small LFP, 2 h	368 372.49	43 412.03	−281 548.43
Highland tea	Overnight withering BESS	860 825.08	590 309.49	319 793.89

Because rejected otherwise-exportable energy is curtailed, modeled NPV falls from full export to partial acceptance and again to no export, turning negative for coffee and pepper while tea remains positive. Export access is thus decision-relevant under the assumptions tested, yet these model runs do not establish that the local grid can accept the modeled energy or deliver the assumed tariff; site-specific interconnection and tariff data are required before the values can be read as project returns.

S7. Device-fabrication tolerance, angular acceptance, and structural-load evidence

S7.1. Thickness tolerance of the focal ST-OPV stack

An independent optical check of the modelled stack cannot be reported here, so instead the model was asked what such a check would have to reproduce. The focal APT44 design ($d_{\mathrm{act}} = 60\,\mathrm{nm}$, $d_{\mathrm{Ag}} = 12\,\mathrm{nm}$, $d_{\mathrm{cap}} = 35\,\mathrm{nm}$, ITO 100 nm, ZnO 35 nm) was re-evaluated with layer thicknesses drawn from illustrative roll-to-roll spreads (active layer 60 nm at 5 % standard deviation, Ag at 1 nm, capping MoO_3 at 2 nm; 4000 draws, seed 42, `scripts/wp2b_device_tolerance_audit.py`). Photon conservation holds to $\max|R + T + A_{\mathrm{tot}} - 1| = 4.44 \times 10^{-16}$ over every draw.

The spread is not uniform across the three figures of merit (Table S9). Ag thickness dominates, moving APT by −3.99 to +3.82 pp at $\pm 2\sigma$; the active layer contributes −2.12 to +2.25 pp and the capping layer at most 0.76 pp. Because PCE and APT move in opposite directions, LUE is comparatively stiff: its standard deviation is 1.3 % of the nominal value. Any ranking that rests on LUE is therefore insensitive to fabrication tolerance, whereas a claim that quotes APT alone inherits a standard deviation of 2.3 pp.

Table S9: Fabrication-tolerance response of the focal APT44 stack under 4000 thickness draws. The standard deviation is across draws, not an experimental repeatability, and the spreads are illustrative rather than measured line statistics. Draws were truncated at 4 nm (Ag) and 0 nm (capping layer) to keep thicknesses physical, so the reported tails are conservative.

Quantity	Nominal	p_{05}	p_{50}	p_{95}	SD
APT (%)	44.12	40.30	44.10	47.75	2.29
PCE (%)	9.93	9.18	9.90	10.61	0.44
LUE (%)	4.38	4.24	4.37	4.43	0.06

S7.2. Angular acceptance

Beam transmittance falls steeply with incidence angle while PCE is nearly flat: APT decreases from 44.12 % at normal incidence to 33.07 % at 60° and 20.00 % at 75°, whereas PCE stays between 9.06 % and 10.19 % over the same range. A gonioscopic measurement, or one reported only at normal incidence, therefore cannot be compared with the diffuse-weighted results used here without declaring its angular protocol; the hemispherical average in Equation (S1) is the quantity that enters the hourly canopy model.

S7.3. Structural and wind-load evidence

The elevated structures require a local wind-load class that this package cannot supply. The shipped ERA5 year contains hourly 10 m means only: annual means of 1.18 to 1.89 $\mathrm{m\,s^{-1}}$, p_{99} of 2.94 to 5.31 $\mathrm{m\,s^{-1}}$, record maxima of 4.58 to 7.03 $\mathrm{m\,s^{-1}}$, and no hour above 10 $\mathrm{m\,s^{-1}}$ at any site (`scripts/wp1_wind_load_screening.py`). Those values correspond to dynamic pressures of 13.1 to 30.9 Pa, one to two orders of magnitude below the 250 to 640 Pa obtained for candidate basic velocities of 20 to 32 $\mathrm{m\,s^{-1}}$. A one-year record of hourly means thus cannot bound a design load, and no design pressure is asserted here. What a local annex must supply is explicit: the national-annex basic velocity, the terrain category, the orography factor, and the altitude correction. Structural cost is already carried in the uncertainty analysis as $C_{\mathrm{BOS}}^{\text{c-Si}} \in [35, 85]$ USD $\mathrm{m^{-2}}$, whose total-order index on comparative parity reaches 0.170, so the economics brackets the structural class while leaving it unvalidated.

S7.4. Field validation and reporting protocol

Paired field validation remains outstanding and is not claimed anywhere in this work. The pre-registered design that would test the reported contrasts is documented in the package (`docs/FIELD_VALIDATION_PROTOCOL_20261114.md`) and fixes, before any data are collected: the four-arm plot layout (open sun, shade tree, shade-matched c-Si, ST-OPV), paired analysis on plot-level differences, primary endpoints (marketable yield, incident and transmitted PAR and spectrum, leaf temperature, VPD, soil moisture), exclusions, and the decision rule. Replication cannot be set from this model, because the modelled inter-year standard deviation of the OPV–c–Si contrast (0.06 to 0.19 pp) reflects shared weather rather than plot variance; the protocol therefore tables replication against assumed plot-level standard deviations instead of asserting a single number.

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